

Manas Dubey

701-138-5250 | manas.dubey007@gmail.com | [LinkedIn](#) | [GitHub](#)

EDUCATION

Manipal University Jaipur

B.Tech Computer Science Engineering (AI & ML)

The Air Force School

CBSE Board Examination (X & XII)

Rajasthan, IN

Aug. 2022 – June 2026

New Delhi, IN

Apr. 2020 – May 2022

EXPERIENCE

NeoFab.AI (AI Factory Planning)

Seattle, WA

Software Engineer Intern (Full-Stack + AI)

Aug. 2025 – Present

- Engineered a Python ETL pipeline (Firecrawl, regex parsing) crawling manufacturer pages into Firestore, scaling the dataset $\sim 17\times$ (**54 to 962** records) across 4 sectors.
- Developed the Next.js + TypeScript landing page with a live Three.js factory simulation, used as the company's investor deck, plus a config system (workflow, process-flow, sidebar editors) on FastAPI endpoints with RBAC.
- Designed a procurement agent that auto-sends requirement-based RFQs to scraped vendor emails and threads replies into an in-dashboard inbox for direct supplier chat.
- Grew a RAG knowledge base to **908** machine records across 4 manufacturing domains; shipped to a live Industrial AI product evaluated by OLA, Ather, and Siemens.

PROJECTS

Legally AI | *Next.js, FastAPI, TypeScript, FAISS, InLegalBERT, Docker*

[Live](#) | [Code](#)

- Architected an AI legal search tool over **88,988** passages from **1,991** Supreme Court judgments, using a multi-stage RAG pipeline (query reformulation, vector search, citation verification) to cut hallucinated citations.
- Shipped a real-time multilingual (Hindi/English/Hinglish) voice assistant (WebRTC + Deepgram STT $\rightarrow$ Groq Llama 3.3 70B $\rightarrow$ TTS) at **<800 ms** P50 latency with interruptible turn-taking and cited summaries.
- Added a reasoning layer combining precedent retrieval, InLegalBERT classification, and grounded LLM inference, returning confidence scores with source-linked evidence.
- Implemented a PDF/DOCX/OCR ingestion pipeline for clause extraction, document chat, and statute identification.

Conduit | *Python, FastAPI, Redis, PostgreSQL, Docker, Prometheus, Grafana*

[Code](#)

- Architected a drop-in OpenAI-compatible LLM gateway over **5** providers; cut API cost **80%** on a **2,000**-request load test via semantic caching and complexity-based routing.
- Engineered a Redis-backed semantic cache (vector similarity, TTL tiers) hitting **70%** cache rate and **77 ms** P95 versus **9.1 s**, a **99%** latency cut.
- Built a fault-tolerant resilience layer (token-bucket rate limiting, budget caps, exponential backoff, circuit breakers, cross-provider failover) with zero caller-visible downtime.
- Instrumented full-path observability (Prometheus, Grafana) and shipped a one-command Docker stack plus a CI/CD pipeline (ruff, mypy, pytest, **70%** coverage gate).

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C++, Java

Frameworks & Web: React, Next.js, FastAPI, Node.js, Express.js, Tailwind CSS

AI/ML: PyTorch, Hugging Face Transformers, FAISS, pgvector, RAG, Vector Search, InLegalBERT, NLP

Data & Backend: ETL Pipelines, REST APIs, PostgreSQL, Redis, Cloud Firestore, MongoDB, Web Scraping, RBAC

DevOps & Tools: Git, Docker, CI/CD, Google Cloud, Firebase, Prometheus, Grafana, pytest, Postman

ACHIEVEMENTS

- Received the Student Excellence Award from the Dept. of AIML, MUJ, and Dean's List Award.
- Won **Rs.12,00,000+** in seed funding from MUJ E-Cell for Legally AI, an affordable legal-research tool for Indian users.
- Published review paper: "*A Comprehensive Review on Object Detection: Advances and Challenges Over 25 Years*".